Setting up the assumptions

- For continuous-time systems we have:

$$\dot{x}(t) = Ax(t) + B_u u(t) + B_w w(t),$$

where A, B_u and B_w are assumed to be known and constant (can generalize later if needed) and:

$$\begin{aligned} x(t) &: \text{ State vector, a random process.} \\ u(t) &: \text{ Deterministic control inputs.} \\ w(t) &: \text{ Noise driving the process (process noise), a random process.} \end{aligned}$$

- Further:

$$\mathbb{E}[x(0)] = \bar{x}(0); \qquad \mathbb{E}[(x(0) - \bar{x}(0))(x(0) - \bar{x}(0))^T] = \Sigma_{\tilde{x}}(0)$$
$$\mathbb{E}[w(t)] = 0; \qquad \mathbb{E}[w(t)w(\tau)^T] = S_w \delta(t - \tau).$$

- S_w is the spectral density of $w(t)$.

Propagation of the mean

- Easiest analysis is to discretize model; then, use results obtained earlier for discrete-time systems letting $\Delta t \to 0$.
- We drop deterministic inputs $u(t)$ from the model for simplicity. (They affect only the mean, and in known ways, and we add their influence back in later.)
- Starting with the discrete case, we have:

$$\begin{aligned} \bar{x}_k &= A_d \bar{x}_{k-1} + B_d u_{k-1} \\ A_d &= e^{A\Delta t} \approx I + A\Delta t + \mathcal{O}(\Delta t^2). \end{aligned}$$

- Let $\Delta t \to 0$ and consider case when $u_k = 0$ for all k.

$$\begin{aligned} \bar{x}_k &= (A\Delta t + I)\bar{x}_{k-1} \\ \frac{\bar{x}_k - \bar{x}_{k-1}}{\Delta t} &= A\bar{x}_{k-1}. \end{aligned}$$

- As $\Delta t \to 0$, $\dot{\bar{x}}(t) = A\bar{x}(t)$, as we might expect.

Summarizing propagation of the mean

- We add back in the influence of $u(t)$, skipping the full derivation here for simplicity.

Overall, the mean propagation for continuous-time dynamic systems having random inputs is:

$$\begin{aligned} \bar{x}(0) &: \text{ Given} \\ \dot{\bar{x}}(t) &= A\bar{x}(t) + B_u u(t). \end{aligned}$$

Simulation of a deterministic system and simulation of the mean value of the state for a stochastic system are treated the same way.

Variations about the mean

- The result for discrete-time systems was:

$$\Sigma_{\tilde{x},k} = A_d \Sigma_{\tilde{x},k-1} A_d^T + \Sigma_{\widetilde{w}}.$$

- But, we need a way to relate discrete $\Sigma_{\widetilde{w}}$ to a continuous spectral density S_w before we can proceed.
- Recall the discrete system response in terms of continuous system matrices:

$$\begin{aligned} x_k &= e^{A\Delta t} x_{k-1} + \int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\tau)} B_w w(\tau)\,\mathrm{d}\tau \\ &= e^{A\Delta t} x_{k-1} + w_{k-1}. \end{aligned}$$

- The integral explicitly accounts for variations in the noise during Δt. We have:

$$w_{k-1} = \int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\tau)} B_w w(\tau)\,\mathrm{d}\tau.$$

Evaluating $\Sigma_{\widetilde{w}}$

- Recall, w_k is discrete white noise having covariance:

$$\mathbb{E}[w_k w_l^T] = \begin{cases} \Sigma_{\widetilde{w}}, & k = l; \\ 0, & k \neq l. \end{cases}$$

- Form outer product using w_{k-1} from prior slide to get equivalent $\Sigma_{\widetilde{w}}$

$$\begin{aligned} \Sigma_{\widetilde{w}} &= \mathbb{E}\left[\left(\int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\tau)} B_w w(\tau)\,\mathrm{d}\tau \right) \left(\int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\gamma)} B_w w(\gamma)\,\mathrm{d}\gamma \right)^T \right] \\ &= \mathbb{E}\left[\int_{(k-1)\Delta t}^{k\Delta t} \int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\tau)} B_w w(\tau) w(\gamma)^T B_w^T e^{A^T(k\Delta t-\gamma)}\,\mathrm{d}\tau\,\mathrm{d}\gamma \right]. \end{aligned}$$

- Big ugly mess but has two saving graces:
  1. Expectation can go inside integrals.
  2. $\mathbb{E}[w(\tau)w(\gamma)^T] = S_w \delta(\tau - \gamma)$ ➠ One of the integrals drops out!

The solution for $\Sigma_{\widetilde{w}}$

- So, we have:

$$\Sigma_{\widetilde{w}} = \int_{(k-1)\Delta t}^{k\Delta t} e^{A(k\Delta t-\tau)} B_w S_w B_w^T e^{A^T(k\Delta t-\tau)}\,\mathrm{d}\tau.$$

- **KEY POINT:** While S_w may have a simple form, $\Sigma_{\widetilde{w}}$ will be a full matrix in general.
- One approach to solving the integral is to approximate.
  - As $\Delta t \to 0$, then $k\Delta t - \tau \to 0$ and $e^{A(k\Delta t-\tau)} \approx I + A(k\Delta t - \tau) + \cdots$
  - That is, $e^{A(k\Delta t-\tau)} \approx I$. Then,

$$\Sigma_{\widetilde{w}} \approx \left(B_w S_w B_w^T\right) \Delta t.$$

- We will see a better method to evaluate $\Sigma_{\widetilde{w}}$ when $\Delta t \neq 0$, but for now we continue with this result to determine the continuous-time system covariance propagation.

The solution for $\Sigma_{\tilde{x},k}$

- We now substitute $\Sigma_{\widetilde{w}} \approx B_w S_w B_w^T \Delta t$ and $A_d \approx (I + A\Delta t)$ into the covariance-propagation equation:

$$\begin{aligned}
\Sigma_{\tilde{x},k} &= A_d \Sigma_{\tilde{x},k-1} A_d^T + \Sigma_{\widetilde{w}} \\
&\approx (I + A\Delta t)\Sigma_{\tilde{x},k-1}(I + A\Delta t)^T + B_w S_w B_w^T \Delta t \\
&= \Sigma_{\tilde{x},k-1} + \Delta t (A\Sigma_{\tilde{x},k-1} + \Sigma_{\tilde{x},k-1}A^T + B_w S_w B_w^T) + \mathcal{O}(\Delta t^2) \\
\frac{\Sigma_{\tilde{x},k} - \Sigma_{\tilde{x},k-1}}{\Delta t} &= A\Sigma_{\tilde{x},k-1} + \Sigma_{\tilde{x},k-1}A^T + B_w S_w B_w^T + \mathcal{O}(\Delta t).
\end{aligned}$$

- As $\Delta t \to 0$

$$\dot{\Sigma}_{\tilde{x}}(t) = A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T,$$

initialized with $\Sigma_{\tilde{x}}(0)$.

Interpreting the solution for $\Sigma_{\tilde{x},k}$

- Covariance: $\dot{\Sigma}_{\tilde{x}}(t) = A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T$.
- This is a matrix differential equation.
- Symmetric, so don't need to solve for every element. Two effects:
  - $A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T$: Homogeneous part. Contractive for stable A. Reduces covariance.
  - $B_w S_w B_w^T$: Impact of process noise. Tends to increase covariance.
- Steady-state solution: Effects balance for systems having constant matrices A, B_w, S_w and stable A.

$$A\Sigma_{\tilde{x},ss} + \Sigma_{\tilde{x},ss}A^T + B_w S_w B_w^T = 0.$$

- This is a continuous-time Lyapunov equation. In Octave, `lyap.m`

Summarizing propagation of the covariance

- So, after a short derivation, we now have the solution for how the uncertainty of the state is propagated over time.

The covariance propagation is:

$$\begin{aligned}
\Sigma_{\tilde{x}}(0) &: \text{ Given} \\
\dot{\Sigma}_{\tilde{x}}(t) &= A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T.
\end{aligned}$$

In this equation, $A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T$ is the homogeneous part; $B_w S_w B_w^T$ is the driving term.

An example, solving for steady-state

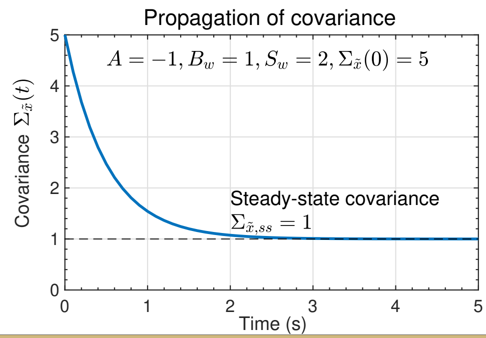
- $\dot{x}(t) = Ax(t) + B_w w(t)$, where A and B_w are scalars.
- Then, $\dot{\Sigma}_{\tilde{x}} = 2A\Sigma_{\tilde{x}} + B_w^2 S_w$; the solution to this ODE is:

$$\Sigma_{\tilde{x}}(t) = \frac{B_w^2 S_w}{2A}(e^{2At} - 1) + \Sigma_{\tilde{x}}(0)e^{2At}.$$

- If $A < 0$ (stable) then the initial condition contribution goes to zero and:

$$\Sigma_{\tilde{x},ss} = \frac{-B_w^2 S_w}{2A}.$$

- Increased $\Sigma_{\tilde{x},ss}$ as more noise added via the $B_w^2 S_w$ term; decreased $\Sigma_{\tilde{x},ss}$ as A becomes "more stable".
- Example shown to the right.



Summary

- With deterministic state-space systems, we can simulate a model and have no uncertainty regarding the state trajectory.
- With stochastic state-space systems, we instead track the mean and covariance of the RV representing the state at every point in time.
- You learned how to update the mean and covariance:

$$\dot{\bar{x}}(t) = A\bar{x}(t) + B_u u(t).$$
$$\dot{\Sigma}_{\tilde{x}}(t) = A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T.$$

- The true state $x(t)$ is expected to be within $\bar{x}(t) \pm 3\sqrt{\mathrm{diag}(\Sigma_{\tilde{x}}(t))}$ approximately 99.7 $\%$ of the time if the model is correct and noises are Gaussian.